Bench exercise: gooseneck and dovetail, side by side

Japanese Joinery by Hand · Episode 9. Draw the koshikake kama-tsugi 腰掛け鎌継ぎ full size in plan next to Episode 5's dovetail. Then build the gooseneck in stiff card so you can drop it in and pull on it. You need card, a craft knife, a steel rule and a square. About 45 minutes.

1 · Draw both plans, full size

1. On paper, draw a 120 mm-wide strip (the gooseneck's timber) with a center line, and a shoulder line across it.
2. From the shoulder, draw a neck 30 mm wide (15 each side of the center line) and 75 mm long. Then draw a head 75 mm long: 45 mm wide at the neck end (7.5 mm more each side), narrowing to about 30 mm at its flat tip.
3. Beside it, draw a 105 mm strip with Episode 5's dovetail: neck 30, length 50, head 50.
4. Compare them. Where would each one bear when pulled?

2 · Build the gooseneck in card

1. Cut the gooseneck outline from two layers of card glued together; this is the male's upper half. Glue it to a card block 120 mm wide.
2. Cut a matching socket out of a second, female block's upper layer, leaving the lower layer as the seat.

3 · Try it

1. Drop the male in from above. It should only go in this way.
2. Pull along the timbers. Feel where the head's flared back bears.
3. Try to lift it. Nothing in the joint stops that.

What to notice

- The gooseneck reaches much further than the dovetail drawing, but its bearing faces are small: only the 7.5 mm flares.
- Stress gathers near the neck, so the neck is the weak point. Researchers found a longer gooseneck isn't simply stronger.
- It holds a pull but barely resists bending. In design it is often treated as a pin.

These sizes are one test series' specimen geometry on 120 mm timbers (denmoku-db; Toda 2012), not a carpentry standard. No carpentry source gives standard sizes. The tip width is read from a specimen drawing; the bearing slope and entry taper are left out.